

GATE EY

Ecology & Evolution

TREND-BASED MOCK PAPER

— 02 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 35 questions (54%)

Difficult: 10 questions (15%)

GENERAL INSTRUCTIONS

GATE EY — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [no -ve]

[General Aptitude]

Antonym of 'indigenous':

Enter your answer here

Q2. 1M [-0.33]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q3. 1M [no -ve]

[Ecology]

Logistic growth has:

Enter your answer here

Q4. 1M [-0.33]

[Evolution]

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

Q5. 1M [no -ve]

[Genetics]

Hardy-Weinberg needs:

Enter your answer here

Q6. 1M [-0.33]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q7. 1M [-0.33]

[Cell Biology]

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

Q8.

1M [-0.33]

[Plant Physiology]

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

Q9.

1M [no -ve]

[Zoology]

Binomial nomenclature by:

Enter your answer here

Q10.

1M [-0.33]

[Behavioral Ecology]

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

Q11.

1M [no -ve]

[Evolution]

Darwin finches from:

Enter your answer here

Q12.

1M [-0.33]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q13.

1M [no -ve]

[Genetics]

Hardy-Weinberg needs:

Enter your answer here

Q14.

1M [no -ve]

[Plant Physiology]

Photosynthesis in:

Enter your answer here

Q15.

1M [no -ve]

[Environmental Science]

Biodiversity hotspot criteria:

*Enter your answer here***Q16.**

1M [-0.33]

[Ecology]

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

Q17.

1M [-0.33]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q18.

1M [no -ve]

[Cell Biology]

Mitochondria function:

*Enter your answer here***Q19.**

1M [-0.33]

[Genetics]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q20.

1M [-0.33]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q21.

1M [-0.33]

[Evolution]

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

Q22.

1M [-0.33]

[Evolution]

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

Q23.

1M [-0.33]

[Behavioral Ecology]

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

Q24.

1M [-0.33]

[Plant Physiology]

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

Q25.

1M [-0.33]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Q26.

1M [-0.33]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q27.

1M [-0.33]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q28.

1M [no -ve]

[Environmental Science]

Biodiversity hotspot criteria:

*Enter your answer here***Q29.**

1M [no -ve]

[Environmental Science]

Biodiversity hotspot criteria:

*Enter your answer here***Q30.**

1M [-0.33]

[Plant Physiology]

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

Q31.

2M [-0.66]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q32.

2M [-0.66]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q33.

2M [no -ve]

[Evolution]

Darwin finches from:

*Enter your answer here***Q34.**

2M [-0.66]

[Cell Biology]

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

Q35.

2M [-0.66]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q36.

2M [-0.66]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Q37.

2M [-0.66]

[Genetics]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q38.

2M [-0.66]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q39.

2M [-0.66]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Q40.

2M [-0.66]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q41.

2M [no -ve]

[General Aptitude]

Antonym of 'indigenous':

*Enter your answer here***Q42.**

2M [no -ve]

[Behavioral Ecology]

Monarch butterfly migrates from:

*Enter your answer here***Q43.**

2M [no -ve]

[General Aptitude]

Antonym of 'indigenous':

*Enter your answer here***Q44.**

2M [-0.66]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q45.

2M [-0.66]

[Environmental Science]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q46.

2M [-0.66]

[Genetics]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q47.

2M [-0.66]

[Behavioral Ecology]

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

Q48.

2M [-0.66]

[Ecology]

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

Q49.

2M [no -ve]

[Behavioral Ecology]

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

Q50.

2M [no -ve]

[Cell Biology]

Mitochondria function:

*Enter your answer here***Q51.**

2M [no -ve]

[Genetics]

Hardy-Weinberg needs:

*Enter your answer here***Q52.**

2M [-0.66]

[Engineering Mathematics]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q53.

2M [-0.66]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Q54.

2M [no -ve]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q55.

2M [no -ve]

[Cell Biology]

Mitochondria function:

*Enter your answer here***Q56.**

2M [-0.66]

[Behavioral Ecology]

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

Q57.

2M [-0.66]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q58.

2M [no -ve]

[General Aptitude]

Antonym of 'indigenous':

*Enter your answer here***Q59.**

2M [no -ve]

[Ecology]

Logistic growth has:

*Enter your answer here***Q60.**

2M [-0.66]

[Zoology]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q61.

2M [-0.66]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Q62.

2M [-0.66]

[Genetics]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q63.

2M [no -ve]

[Cell Biology]

Mitochondria function:

*Enter your answer here***Q64.**

2M [-0.66]

[Genetics]

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

Q65.

2M [-0.66]

[General Aptitude]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original